

MailFlow: System Architecture & Specification

Engineering Reference Document for Antigravity & AI Development

1. Executive Summary & Application Scope

MailFlow is a modern, full-stack Gmail intelligence and bulk management platform designed to automate high-volume inbox decluttering, execute precision batch actions, and reveal behavioral patterns across recurring, automated, and unread communications.

2. Technical Stack & Infrastructure

- 🔧 Frontend: React 19, TypeScript, Vite 6, Tailwind CSS v4, Motion (transitions), Lucide React.
- 🔧 Backend API Gateway: Node.js / Express 4.x compiled to standalone CommonJS (dist/server.cjs) via esbuild.
- 🔧 Authentication: Firebase Authentication + Google Identity Services / OAuth 2.0 (scope: `https://mail.google.com/`).
- 🔧 AI Engine: Multi-provider LLM gateway supporting Google Gemini (@google/genai), OpenAI, Anthropic, DeepSeek, Groq, Mistral, and Zhipu AI.

3. Core Functional Modules

- 🔧 Natural Language Smart Search: Converts free-form input into strict Gmail search query operators (e.g. `larger:5M, category:promotions, before:YYYY/MM/DD, is:unread`).
- 🔧 Inbox Health & Pattern Clustering: Samples recent headers to identify automated behavioral bundles (e.g., Ignored Blasts, Financial Summaries, Receipts, Trial Expirations) with 1-click bulk purge.
- 🔧 High-Throughput Batch Processing: Processes operations in concurrent chunks (15-20 messages per batch) to avoid Google API throttling: Trash, Archive, Mark as Read, Permanent Delete.
- 🔧 Exact Telemetry & Real-Time Aggregation: Performs live whole-inbox sweeps (`countEmails`) for top senders and domains to eliminate sampling discrepancies.

4. Server Endpoints & Data Contracts

- 🔧 POST `/api/parse-query`: Input: { query, settings } -> Output: { query: string, suggestedFolder?: string, explanation?: string }
- 🔧 POST `/api/analyze-inbox`: Input: { emails: Array<{id, sender, subject, labelIds}>, settings } -> Output: { clusters: Array<{title, description, suggestedAction, searchQuery, estimatedCount, patternDetected}> }
- 🔧 POST `/api/models`: Input: { provider, apiKey, baseUrl? } -> Output: { models: string[] } for dynamic model selection.

5. Resilience, Error Handling & Quota Protection

The platform features automated error interception for AI services:

- 🔧 HTTP 429 Quota Exceeded: Intercepted gracefully; disables AI for the current search, activates standard Gmail fallback, and prompts the user to enter their personal API key.
- 🔧 HTTP 503 Service Unavailable: Detects provider overload and recommends switching models or providers in Settings without crashing the client UI.

6. Antigravity Developer Roadmap (Next Priorities)

- 🔧 Phase 1: Pagination & Deep Loading: Implement cursor-based pagination with `nextPageToken` when search yields >100 items, showing total matching count.
- 🔧 Phase 2: Interactive Distribution Modal: Provide an on-demand modal chart visualizing category splits (Primary, Promotions, Social, Updates) and read vs unread ratios.

ðŸŒŠ Phase 3: Automated Rule Creation: Enable 1-click conversion of identified AI clusters into persistent Gmail filters (users.me.settings.filters).

ðŸŒŠ **Phase 4: Audit & Export Logs: Offer CSV/JSON export of message metadata prior to permanent batch deletion.**